

Work Done and Power

SS ILP Physics (Intermediate), Week 3 Class 3



Outline

- Definition of work done
- Recap: dot product
- Work done by multiple external forces
- Definition of power

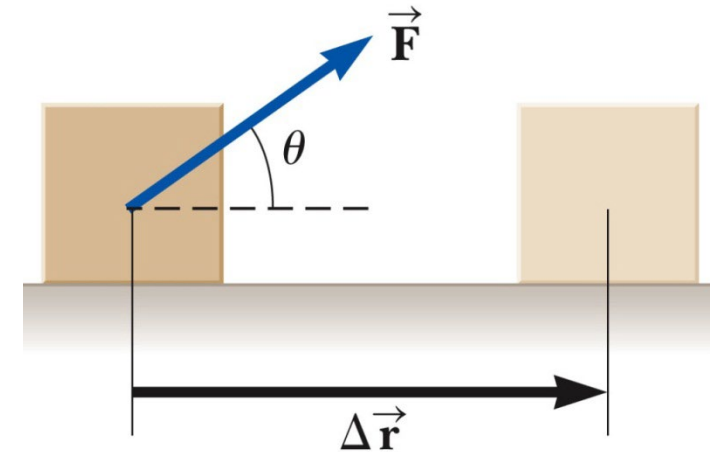
What is the meaning of work done?

- **Work done**, W , on a system by a constant force on the system is the product of the magnitude F of the force, the magnitude Δr of the displacement of the point of application of the force, and $\cos \theta$, where θ is the angle between the force and the displacement vectors (ie resolving displacement in the direction of the force).
- $W = F \Delta r \cos \theta$

Definition of work done: Dot product

- $W = F \Delta r \cos \theta$
 - The displacement is that of the point of application of the force.
 - A force does no work on the object if the force does not move through a displacement.
 - The work done by a force on a moving object is zero when the force applied is perpendicular to the displacement of its point of application.
- Mathematically, it is a dot product:

$$\text{work done} = \vec{F} \cdot \Delta \vec{r}$$



R.A. Serway and J.W. Jewett, Physics for Scientists and Engineers with Modern Physics. 9E, Cengage Learning, Boston, 2014.

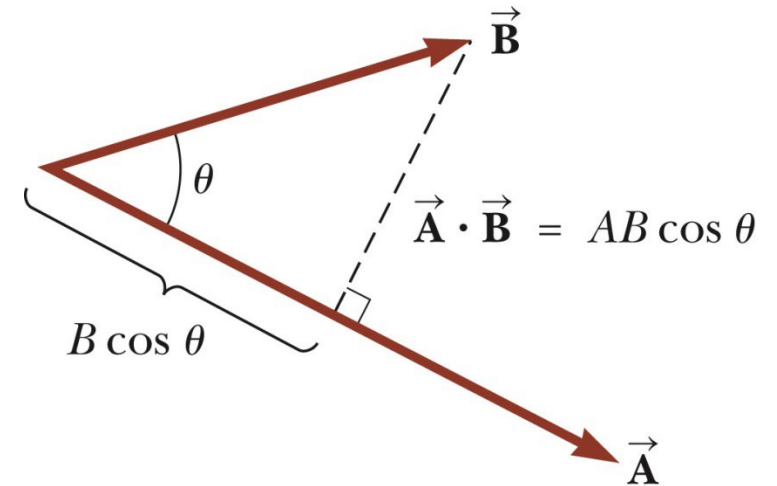
Re-cap: Scalar product of Two Vectors

- The scalar product of two vectors is written as $\vec{A} \cdot \vec{B}$.
 - It is also called the dot product.

- $\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta$
 - θ is the angle *between* A and B

- Applied to work, this means

$$W = F \Delta r \cos \theta = \vec{F} \cdot \Delta \vec{r}$$



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Re-cap: Dot Products

- Dot product of unit vectors $\hat{\mathbf{i}} \cdot \hat{\mathbf{i}} = \hat{\mathbf{j}} \cdot \hat{\mathbf{j}} = \hat{\mathbf{k}} \cdot \hat{\mathbf{k}} = 1$
 $\hat{\mathbf{i}} \cdot \hat{\mathbf{j}} = \hat{\mathbf{j}} \cdot \hat{\mathbf{k}} = \hat{\mathbf{i}} \cdot \hat{\mathbf{k}} = 0$

- Using component form with vectors

$$\vec{A} = A_x \hat{\mathbf{i}} + A_y \hat{\mathbf{j}} + A_z \hat{\mathbf{k}}$$

$$\vec{B} = B_x \hat{\mathbf{i}} + B_y \hat{\mathbf{j}} + B_z \hat{\mathbf{k}}$$

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

- In the special case where $\vec{A} = \vec{B}$

$$\vec{A} \cdot \vec{B} = A_x A_x + A_y A_y + A_z A_z = A^2$$

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Re-cap: Dot Products

- Properties:

- The scalar product is commutative.

$$\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$$

- The scalar product obeys the distributive law of multiplication.

$$\vec{A} \cdot (\vec{B} + \vec{C}) = \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C}$$

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Work done, cont.

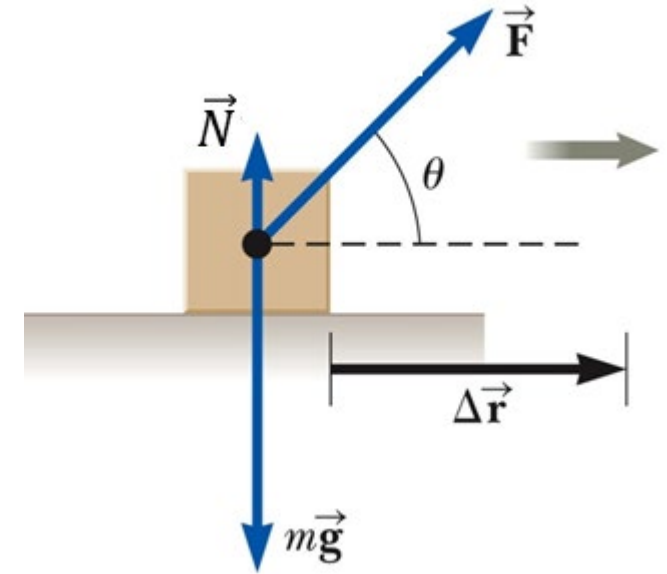
- Force is said to do work when it acts on a body so that there is a displacement of the point of application
- Work done by the force will result in a transfer of energy from/to the system that the force is acting on
- How do we approach problems dealing with forces?
 - Draw free body diagrams
 - Enable you to have a clearer idea of the forces

Concept Questions

The figure on right shows a force $\vec{F}$ acting on an object moving along the horizontal surface.

1. Does the normal reaction force $\vec{N}$ do any work?
2. Does the gravitational force do any work?
3. What is the work done by the force $\vec{F}$?

The force $\vec{F}$ is the only force that does work in this example



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More about work

- The sign of the work depends on the direction of the force relative to the displacement.
 - Work is positive when $0 \leq \theta < \pi/2$.
 - Work is negative when $\pi/2 < \theta \leq \pi$.
- The work done by a force can be calculated, but that force is not necessarily the cause of the displacement.
- Work is a scalar quantity.
- The unit of work is a joule (J)
 - $1 \text{ joule} = 1 \text{ newton} \cdot 1 \text{ meter} = \text{kg} \cdot \text{m}^2 / \text{s}^2$
 - $\text{J} = \text{N} \cdot \text{m} = \text{kg} \cdot \text{m}^2 / \text{s}^2$

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Work Is An Energy Transfer

- If a system interacts with its environment, this interaction can be described as a transfer of energy across the system boundary. This will result in a change in the amount of energy stored in the system.

$$W = E_f - E_i$$

- If the work is done on a system and it is positive, energy is transferred to the system.
- If the work done on the system is negative, energy is transferred from the system.

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Work Done By Multiple Forces

- If more than one force acts on a system and the system can be modeled as a particle, the total work done on the system is the work done by the net force.

$$W_{total} = \sum_i W_{ext,i} = \sum_i \vec{F}_i \cdot \Delta \vec{r}$$

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Activity 1

- (a) Steve exerts a steady force of magnitude 210 N on the stalled car as he pushes it a distance of 18 m. The car also has a flat tire, so to make the car track straight, Steve must push at an angle of 30° to the direction of motion. How much work does Steve do?
- (b) In a helpful mood, Steve pushes a second stalled car with a steady force, $\vec{F} = (160 \text{ N})\hat{i} - (40 \text{ N})\hat{j}$. The displacement of the car is $\Delta\vec{r} = (14 \text{ m})\hat{i} - (11 \text{ m})\hat{j}$. How much work does Steve do in this case?

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Activity 1 (Solution)

(a) Work done by Steve is given by

$$W = \vec{F} \cdot \Delta\vec{r} = F\Delta r \cos \theta = 210 \times 18 \times \cos 30^\circ \text{ J} = 3.3 \times 10^3 \text{ J}$$

(b) Work done by Steve is given by

$$W = \vec{F} \cdot \Delta\vec{r} = [(160 \text{ N})\hat{i} - (40 \text{ N})\hat{j}] \cdot [(14 \text{ m})\hat{i} - (11 \text{ m})\hat{j}]$$

$$W = (160 \cdot 14 \text{ J}) + (40 \cdot 11 \text{ J}) = 2.68 \times 10^3 \text{ J}$$

Activity 2

You push your laptop 1.50 m along a horizontal tabletop with a horizontal push of 2.40 N while the opposing force of friction produced by the tabletop surface is 0.600 N. How much work does each of the following forces do on the book:

- (a) your 2.40 N push;
- (b) the friction force;
- (c) the normal force from the tabletop; and
- (d) gravity?

What is the net work done on the book?

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Activity 2 (Solution)

(a) Work done by the “2.40 N” force:

$$W_F = F\Delta r \cos \theta = 2.40 \times 1.50 \times \cos 0^\circ = 3.60 \text{ J}$$

(b) Work done by the friction force, f :

$$W_f = f\Delta r \cos \theta = 0.600 \times 1.50 \times \cos 180^\circ = -0.90 \text{ J}$$

(c) Work done by the normal force, N :

$$W_N = N\Delta r \cos \theta = Ns \times \cos 90^\circ = 0 \text{ J}$$

(d) Work done by the gravity, mg :

$$W_{grav} = mg\Delta r \cos \theta = mg \cos 90^\circ = 0 \text{ J}$$

Total work done:

$$W = W_F + W_f + W_N + W_{grav} = 2.70 \text{ J}$$

Activity 3

A tow truck pulls a car 5.00 km along a horizontal roadway using a cable having a tension of 850 N.

- (a) How much work does the cable do on the car if it pulls horizontally? If it pulls 35 degree at above the horizontal?
- (b) How much work does the cable do on the tow truck in both cases of part (a)?
- (c) How much work does gravity do on the car in part (a)?

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Activity 3 (Solution)

- (a) When the cable pulls the car horizontally, $\theta = 0^\circ$. Work done on the car by the rope is:

$$W = F\Delta r \cos \theta = 850 \text{ N} \cdot 5000 \text{ m} \cdot \cos 0^\circ = 4.26 \times 10^6 \text{ J}$$

When the cable pulls the car at an angle of $\theta = 35^\circ$, the work done on the car by the rope becomes

$$W = F\Delta r \cos \theta = 850 \text{ N} \cdot 5000 \text{ m} \cdot \cos 35^\circ = 3.48 \times 10^6 \text{ J}$$

Activity 3 (Solution)

- (b) When the cable pulls the car horizontally, the cable exerts an oppositely directed force on the tow truck. In this case, $\theta = 180^\circ$ and the **work done on the tow truck** by the rope is

$$W = F\Delta r \cos \theta = 850 \text{ N} \cdot 5000 \text{ m} \cdot \cos 180^\circ = -4.26 \times 10^6 \text{ J}$$

Similarly, when the cable pulls the car at an angle of 35 degree above the horizon, the cable exerts a force on the tow truck at an angle of 35 degree below the horizon, i.e. $\theta = 90^\circ + 55^\circ = 145^\circ$. The **work done on the tow truck** by the rope is

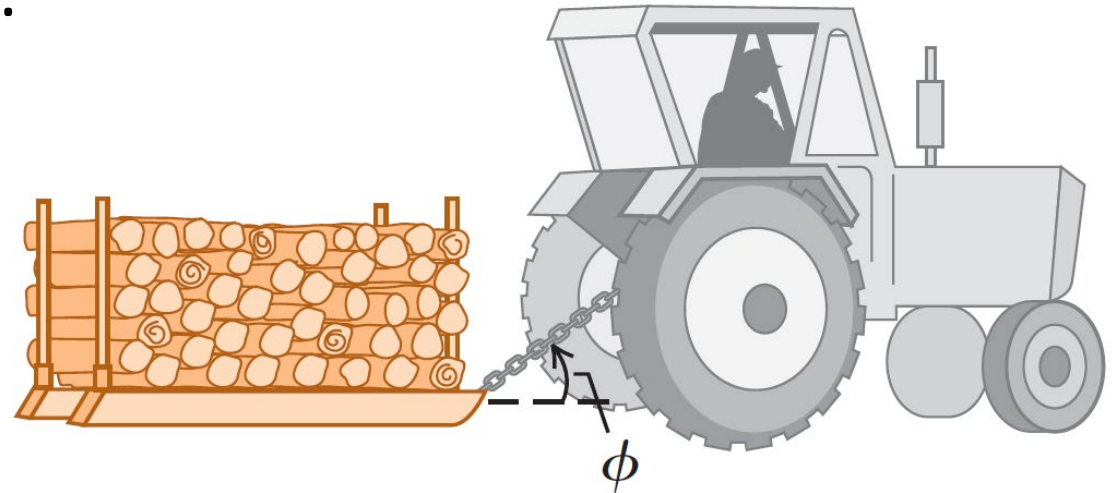
$$W = F\Delta r \cos \theta = 850 \text{ N} \cdot 5000 \text{ m} \cdot \cos 145^\circ = -3.48 \times 10^6 \text{ J}$$

Activity 3 (Solution)

- (c) Gravity force on the car is always directed vertically downwards while the displacement of the car is horizontal. In this case, $\theta = 90^\circ$, and the work done is $W = 0$ J.

Activity 4

A farmer hitches her tractor to a sled loaded with firewood and pulls it a distance of 20 m along level ground (see figure below). The total weight of sled and load is 14,700 N. The tractor exerts a constant 5000-N force at an angle of 36.9 degree above the horizontal. A 3500-N friction force opposes the sled's motion. Find the work done by each force acting on the sled and the total work done by all the forces.

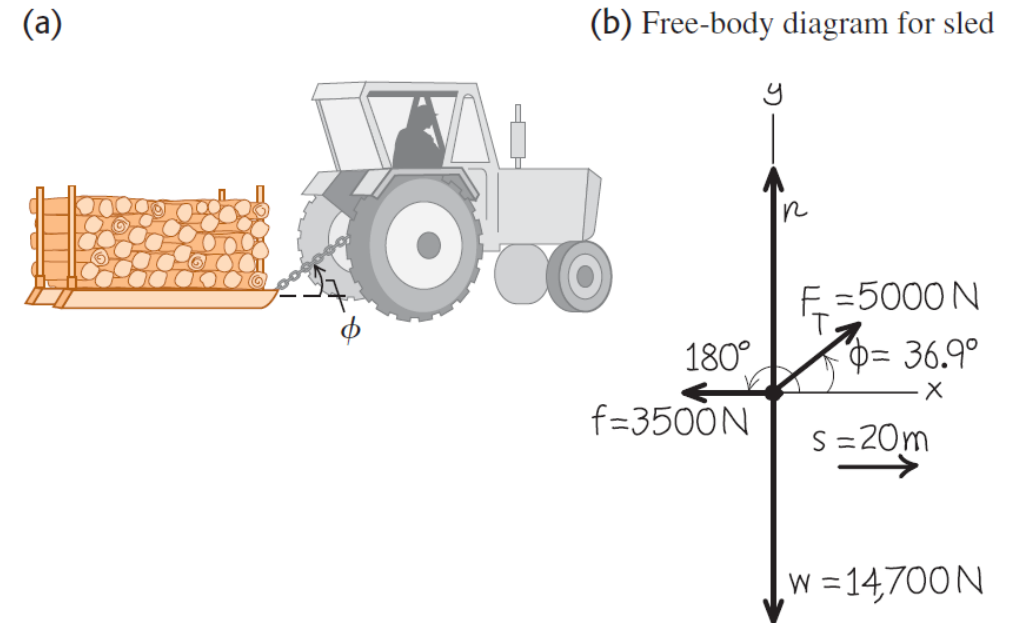


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Activity 4 (Solution)

From the free body diagram, there are four forces acting on the sled:
 (1) normal force from the ground, N ; (2) gravity, w ; (3) friction, f ;
 and (4) pulling force exerted by the tractor, F_T .

Forces (1) and (2) are always
 perpendicular to the displacement.
 Hence, $W_N = W_{grav} = 0 \text{ J}$



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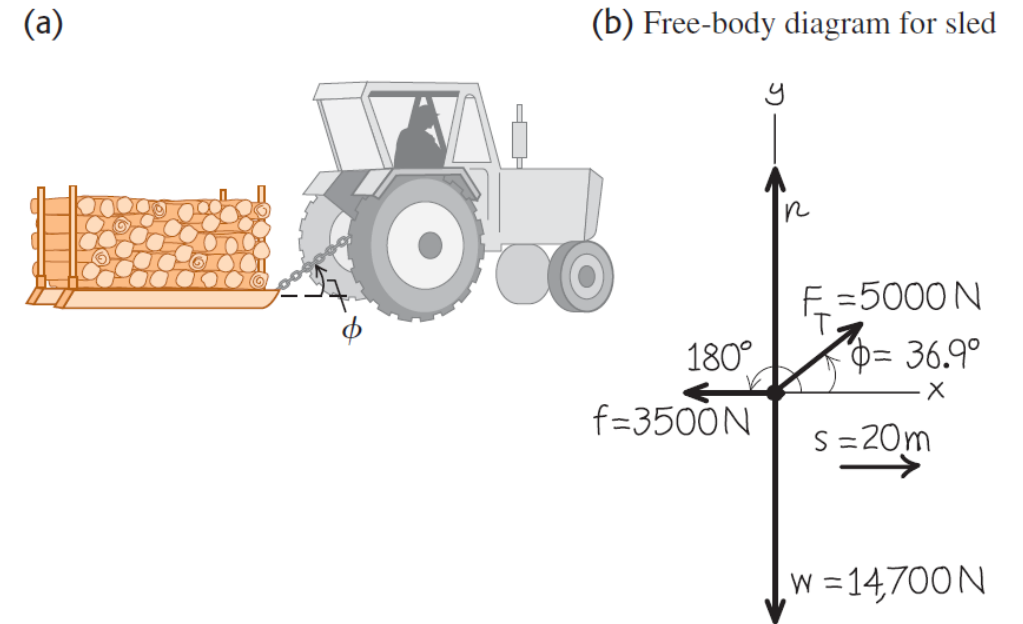
Activity 4 (Solution)

From the free body diagram, there are four forces acting on the sled:
 (1) normal force from the ground, n ; (2) gravity, w ; (3) friction, f ; and
 (4) pulling force exerted by the tractor, F_T .

Angle between force (3) and displacement is $\theta = 180^\circ$,

$$W_f = f \Delta r \cos 180^\circ$$

$$= 3500 \text{ N} \cdot 20 \text{ m} \cdot (-1) = -7 \times 10^4 \text{ J}$$



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Activity 4 (Solution)

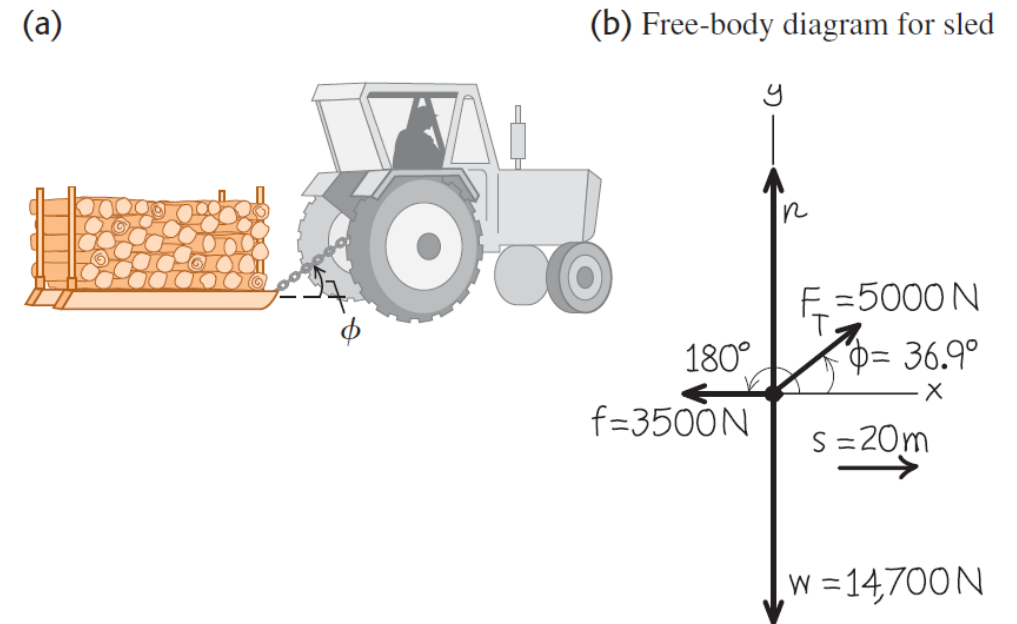
From the free body diagram, there are four forces acting on the sled:
 (1) normal force from the ground, n ; (2) gravity, w ; (3) friction, f ; and
 (4) pulling force exerted by the tractor, F_T .

Angle between force (4) and displacement is $\theta = 36.9^\circ$,

$$\begin{aligned} W_T &= F_T \Delta r \cos \theta \\ &= 5000 \text{ N} \cdot 20 \text{ m} \cdot \cos 36.9^\circ \\ &= 8 \times 10^4 \text{ J} \end{aligned}$$

Total work done on the sled by all forces,

$$W_{total} = W_n + W_{grav} + W_f + W_T = 10^4 \text{ J}$$



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Activity 5

A 75.0-kg painter climbs a ladder that is 2.75 m long leaning against a vertical wall. The ladder makes a 30 degree angle with the wall.

- (a) How much work does gravity do on the painter? Take $g = 9.8\text{m/s}^2$.
- (b) Does the answer to part (a) depend on whether the painter climbs at constant speed or accelerates up the ladder?

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Activity 5 (Solution)

- (a) The gravity force, w , is vertically downward, while the displacement along the ladder, s , makes an angle of 30 degree with the wall. The angle between w and s is

$$\theta = 180^\circ - 30^\circ = 150^\circ$$

Work done by gravity on the painter:

$$\begin{aligned} W &= F\Delta r \cos \theta = w\Delta r \cos \theta = 75 \text{ kg} \cdot 9.8 \text{ m/s}^2 \cdot 2.75 \text{ m} \cdot \cos 150^\circ \\ &= -1750 \text{ J} \end{aligned}$$

- (b) Work done only depends on the dot product between force and displacement vectors, and is independent of the motion of the climber.

Power

- Power is the time rate at which the work is done.
- A quantity of work done ΔW during a time interval Δt , the **average power** is:

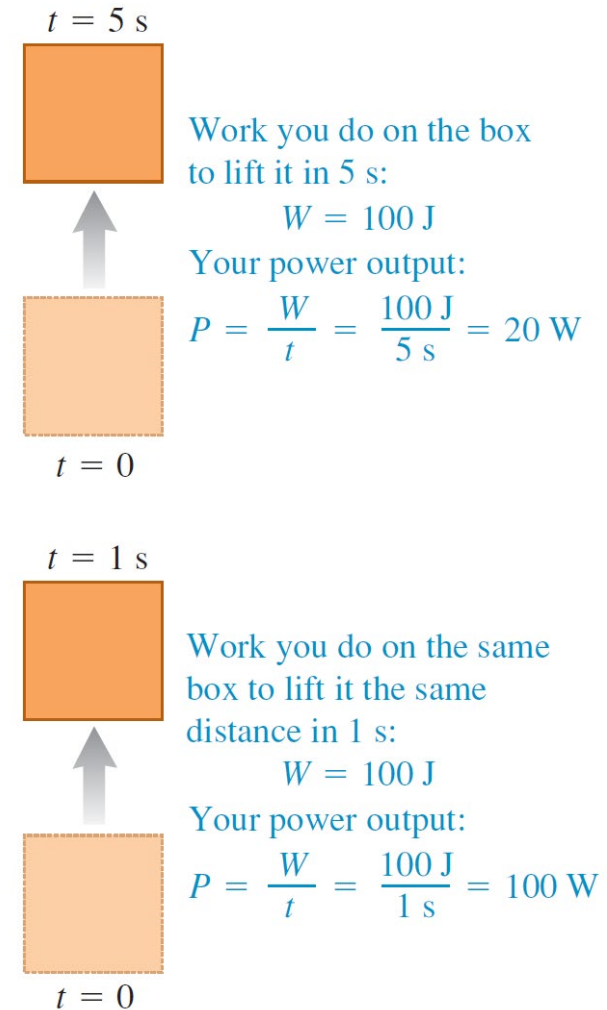
$$P_{ave} = \frac{\Delta W}{\Delta t}$$

- When $\Delta t \rightarrow 0$, the **instantaneous power** is:

$$P = \lim_{\Delta t \rightarrow 0} \frac{\Delta W}{\Delta t} = \frac{dW}{dt}$$

- SI unit: watt (W)
 - $1 \text{ W} = 1 \text{ J/s}$
 - Think:** What does it mean by a “10 W” LED light bulb?
- Commercial unit of electrical energy: kilowatt-hour (kW · h) – unit of energy (work done) not power.

$$1 \text{ kW} \cdot \text{h} = 10^3 \text{ J/s} \times 3600 \text{ s} = 3.6 \text{ MJ}$$



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Power: Force and Velocity

- Work done by a force $\vec{F}$ in displacing a body by a displacement of $\Delta\vec{r}$ is $\Delta W = \vec{F} \cdot \Delta\vec{r}$.
- From the definition of average power:

$$P_{ave} = \frac{\Delta W}{\Delta t} = \frac{\vec{F} \cdot \Delta\vec{r}}{\Delta t} = \vec{F} \cdot \vec{v}_{ave}$$

- In the limit of $\Delta t \rightarrow 0$, the instantaneous power becomes:

$$\boxed{P = \vec{F} \cdot \vec{v}}$$

i.e. the instantaneous power is equal to the dot product between the applied force and the instantaneous velocity of the object

- For linear motion of an object with speed v , the instantaneous power is

$$P = F_{\parallel} v$$

where $F_{\parallel}$ is the force component acting in the same direction as the motion of the object.

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Activity 6

Each of the four jet engines on an Airbus A380 airliner develops a thrust (a forward force on the airliner) of 322,000 N. When the airplane is flying at 900 km/h, what *horsepower* does each engine develop? (Note: 1 horsepower = 746 W)

Activity 6 (Solution)

The thrust is in the direction of motion, so $F_{\parallel}$ is just equal to the thrust. At the power developed by each engine is

$$P = F_{\parallel} v = 3.22 \times 10^5 \text{ N} \times 900 \times \left(\frac{1000}{3600} \right) \text{ ms}^{-1} = 8.05 \times 10^7 \text{ W}$$

Since 1 hp equals to 746 W,

$$P = 8.05 \times 10^7 \text{ W} \times \left(\frac{1 \text{ hp}}{746 \text{ W}} \right) = 1.08 \times 10^5 \text{ hp}$$

Off-class practice question(s)

Activity

A factory worker pushes a 30.0-kg crate a distance of 4.5 m along a level floor at constant velocity by pushing horizontally on it. The coefficient of kinetic friction between the crate and the floor is 0.25.

- (a) What magnitude of force must the worker apply?
- (b) How much work is done on the crate by this force?
- (c) How much work is done on the crate by friction?
- (d) How much work is done on the crate by the normal force? By gravity?
- (e) What is the total work done on the crate?

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Activity (Solution)

- (a) The factory worker must at least exert a force equals to the friction, $f_k = \mu_k N$, where $N = mg$ is the normal force. The magnitude of the force the worker must apply is hence

$$F_{worker} = f_k = \mu_k mg = 0.25 \cdot 30.0 \text{ kg} \cdot 9.80 \text{ ms}^{-2} = 74 \text{ N}$$

- (b) Since the force applied by the worker is horizontal and in the direction of the displacement, $\phi = 0^\circ$ and the work is:

$$W_{worker} = F_{worker} \Delta r \cos 0^\circ = 74 \text{ N} \cdot 4.5 \text{ m} \cdot 1 = 333 \text{ J}$$

- (c) Friction acts in the direction opposite of motion, thus $\phi = 180^\circ$ and the work of friction is:

$$W_{friction} = f_k \Delta r \cos 180^\circ = 74 \text{ N} \cdot 4.5 \text{ m} \cdot (-1) = -333 \text{ J}$$

Activity (Solution)

(d) Both gravity and the normal force act perpendicular to the direction of displacement. Thus, neither force does any work on the crate and $W_{grav} = W_N = 0 \text{ J}$.

(e) The net work done is

$$W_{total} = W_{worker} + W_{friction} + W_{grav} + W_N = 0 \text{ J}$$

The net work done is zero since there is no net force acting on the crate, i.e. the force by the worker exactly cancels the friction force.

Activity

It is 5.0 km from your home to the SUTD. As part of your physical fitness program, you could run that distance at 10 km/h (which uses up energy at the rate of 700 W), or you could walk it leisurely at 3.0 km/h (which uses energy at 290 W). Which choice would burn up more energy, and how much energy (in joules) would it burn? Why is it that the more intense exercise actually burns up less energy than the less intense exercise?

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Activity (Solution)

Time taken for **running** from SUTD to home:

$$\Delta t = \frac{s}{v} = \frac{5.0}{10} \text{ h} = 0.50 \text{ h} = 1800 \text{ s}$$

The energy used is equal to the work done, which is related to the power by,

$$W = P \cdot \Delta t = 700 \text{ W} \cdot 1800 \text{ s} = 1.26 \times 10^6 \text{ J}$$

Time taken for **walking** from SUTD to home:

$$\Delta t = \frac{s}{v} = \frac{5.0}{3} \text{ h} = \frac{5.0}{3} \times 3600 \text{ s} = 6000 \text{ s}$$

The energy used is equal to the work done, which is related to the power by,

$$W = P \cdot \Delta t = 290 \text{ W} \cdot 6000 \text{ s} = 1.74 \times 10^6 \text{ J}$$

The less intense exercise lasts longer and therefore burns up more energy than the intense exercise.